

SIMATS ENGINEERING

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Machine learning

Experiment 1

Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

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# FIND-S Algorithm Implementation

import pandas as pd

# Step 1: Create Training Dataset
data = {
    'Sky': ['Sunny', 'Sunny', 'Rainy', 'Sunny'],
    'AirTemp': ['Warm', 'Warm', 'Cold', 'Warm'],
    'Humidity': ['Normal', 'High', 'High', 'High'],
    'Wind': ['Strong', 'Strong', 'Strong', 'Strong'],
    'Water': ['Warm', 'Warm', 'Warm', 'Cool'],
    'Forecast': ['Same', 'Same', 'Change', 'Same'],
    'EnjoySport': ['Yes', 'Yes', 'No', 'Yes']
}

df = pd.DataFrame(data)

print("Training Data:\n")
print(df)

# Step 2: Separate Features and Target
concepts = df.iloc[:, :-1].values
target = df.iloc[:, -1].values

# Step 3: FIND-S Algorithm
def findS(concepts, target):
    # Initialize hypothesis with the first positive example
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hypothesis = None

for i in range(len(target)):
    if target[i] == "Yes":
        hypothesis = concepts[i].copy()
        break

# Update hypothesis using remaining positive examples
for i in range(len(target)):
    if target[i] == "Yes":
        for j in range(len(hypothesis)):
            if hypothesis[j] != concepts[i][j]:
                hypothesis[j] = '?'

return hypothesis

# Step 4: Find Most Specific Hypothesis
final_hypothesis = findS(concepts, target)

print("\nMost Specific Hypothesis:")
print(final_hypothesis)

```

OUTPUT :

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... Training Data:

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	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
0	Sunny	Warm	Normal	Strong	Warm	Same	Yes
1	Sunny	Warm	High	Strong	Warm	Same	Yes
2	Rainy	Cold	High	Strong	Warm	Change	No
3	Sunny	Warm	High	Strong	Cool	Same	Yes

```

Most Specific Hypothesis:
['Sunny' 'Warm' '?' 'Strong' '?' 'Same']

```